



3 Complex Analysis 2024

Tutorial 3

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Mean value theorem

Problem 11 Consider the function $f(z) = \frac{1}{z}$ which is holomorphic in $\mathbb{C} \setminus \{0\}$. Take $z_0 = 1$, $z_1 = 1 + i$.

- (a) Calculate $\frac{f(z_1) - f(z_0)}{z_1 - z_0}$.
- (b) Show that there is no $c \in [z_0, z_1]$ with $f'(c) = \frac{f(z_1) - f(z_0)}{z_1 - z_0}$.

Solution.

- (a) We have $f(1) = 1$,

$$f(1 + i) = \frac{1}{1 + i} = \frac{1 - i}{(1 + i)(1 - i)} = \frac{1}{2} - \frac{1}{2}i,$$

and

$$\frac{f(1 + i) - f(1)}{(1 + i) - 1} = \frac{\frac{1}{2} - \frac{1}{2}i - 1}{i} = -\frac{1}{2} + \frac{1}{2}i.$$

- (b) We have $f'(z) = -\frac{1}{z^2}$, so we are looking for z with

$$-\frac{1}{z^2} = -\frac{1}{2} + \frac{1}{2}i,$$

i.e.

$$z^2 = -\frac{1}{\left(-\frac{1}{2} + \frac{1}{2}i\right)} = 1 + i.$$

The segment $[1, 1 + i]$ can be parameterized as

$$[1, 1 + i] = \{1 \cdot (1 - t) + (1 + i) \cdot t : 0 \leq t \leq 1\} = \{1 + it : 0 \leq t \leq 1\}.$$

Thus, if $z \in [1, 1 + i]$, then $z^2 = (1 + it)^2 = (1 - t^2) + i2t$ with some $t \in [0, 1]$.
Equation

$$(1 - t^2) + i2t = 1 + i$$

is equivalent to the system of equations

$$1 - t^2 = 1, \quad 2t = 1,$$

which has no solution.

Series

Problem 12 Prove that if $\sum_{k=1}^{\infty} a_k$ is a convergent series of complex numbers, then $\lim_{k \rightarrow \infty} a_k = 0$.

Solution.

Let $s = \sum_{k=1}^{\infty} a_k = \lim_{n \rightarrow \infty} s_n$, where $s_n = \sum_{k=1}^n a_k$. We have

$$a_n = s_n - s_{n-1} \rightarrow s - s = 0, \quad n \rightarrow \infty.$$

Problem 13

- (a) Prove the Comparison Test for complex series: Let $\sum_{k=1}^{\infty} a_k$ be a complex series. Assume that for each $k \in \mathbb{N}$ there is a real number $c_k \geq 0$ such that $|a_k| \leq c_k$. If $\sum_{k=1}^{\infty} c_k < \infty$, then $\sum_{k=1}^{\infty} a_k$ converges.
- (b) Prove that if a complex series $\sum_{k=1}^{\infty} a_k$ converges absolutely, then it converges.

Solution.

- (a) Put $s_n = \sum_{k=1}^n a_k$, then $\operatorname{Re} s_n = \sum_{k=1}^n \operatorname{Re} a_k$, $\operatorname{Im} s_n = \sum_{k=1}^n \operatorname{Im} a_k$. By Lemma 0.2, the sequence $(s_n)_{n=1}^{\infty}$ converges if and only if both sequences $(\operatorname{Re} s_n)_{n=1}^{\infty}$ and $(\operatorname{Im} s_n)_{n=1}^{\infty}$ converge. In other words, the series $\sum_{k=1}^{\infty} a_k$ converges if and only if both real series $\sum_{k=1}^{\infty} \operatorname{Re} a_k$ and $\sum_{k=1}^{\infty} \operatorname{Im} a_k$ converge.

Since $|\operatorname{Re} a_k| \leq |a_k| \leq c_k$, $|\operatorname{Im} a_k| \leq |a_k| \leq c_k$ for all $k \in \mathbb{N}$, both series $\sum_{k=1}^{\infty} \operatorname{Re} a_k$ and $\sum_{k=1}^{\infty} \operatorname{Im} a_k$ converge by the Comparison Test for real series.

- (b) Apply the Comparison Test with $c_k = |a_k|$, $k \in \mathbb{N}$.

Power series

Problem 14 Use the Weierstrass M-Test to show that the series $\sum_{k=1}^{\infty} \frac{z^k}{k^k}$ converges uniformly on any disc of the form $\{z \in \mathbb{C} : |z| \leq r\}$, $r > 0$.

Solution.

Take $N \in \mathbb{N}$, $N > r$. Then for $k \geq N$ we have

$$\left| \frac{z^k}{k^k} \right| = \left(\frac{|z|}{k} \right)^k \leq \left(\frac{r}{N} \right)^k.$$

Since $\sum_{k=1}^{\infty} \left(\frac{r}{N} \right)^k$ is a convergent geometric series, the Weierstrass M-Test implies that $\sum_{k=1}^{\infty} \frac{z^k}{k^k}$ converges uniformly on $\{z \in \mathbb{C} : |z| \leq r\}$.

Problem 15 We know that the series $\sum_{k=0}^{\infty} z^k$ converges uniformly on any disc of the form $\{z \in \mathbb{C} : |z| \leq r\}$, $0 < r < 1$.

- (a) Show this directly using the definition of uniform convergence.
- (b) Show that the series does not converge uniformly on the disc $\{z \in \mathbb{C} : |z| < 1\}$.

Solution.

We know that the series $\sum_{k=0}^{\infty} z^k$ converges for $|z| < 1$, and its sum is $f(z) = \frac{1}{1-z}$.

- (a) We have

$$\left| \sum_{k=0}^n z^k - \frac{1}{1-z} \right| = \left| \frac{1-z^{n+1}}{1-z} - \frac{1}{1-z} \right| = \frac{|z|^{n+1}}{|1-z|}.$$

Let $|z| \leq r$ with $0 < r < 1$. Then $|1-z| \geq 1-r$, this follows from the triangle inequality $1 \leq |z| + |1-z|$. Thus,

$$\left| \sum_{k=0}^n z^k - \frac{1}{1-z} \right| \leq \frac{r^{n+1}}{1-r} \rightarrow 0, \quad n \rightarrow \infty,$$

and the bound does not depend on z . It follows that the convergence is uniform in the disc $\{z \in \mathbb{C} : |z| \leq r\}$.

- (b) Now let us show that the convergence is not uniform in the disc $\{z \in \mathbb{C} : |z| < 1\}$. Take $z_m = 1 - \frac{1}{m}$, $m \in \mathbb{N}$. Then

$$\left| \sum_{k=0}^n z_m^k - \frac{1}{1-z_m} \right| = \frac{|z_m|^{n+1}}{|1-z_m|} = m \left(1 - \frac{1}{m}\right)^{n+1}.$$

For a fixed $n \in \mathbb{N}$ we have

$$\lim_{m \rightarrow \infty} \left| \sum_{k=0}^n z_m^k - \frac{1}{1-z_m} \right| = \infty.$$

Take, for example, $\varepsilon = 1$. The above formula implies that for any $N \in \mathbb{N}$ we can find $n \geq N$ (for example, we take $n = N$) and $m \in \mathbb{N}$ such that

$$\left| \sum_{k=0}^n z_m^k - \frac{1}{1-z_m} \right| \geq 1.$$

Thus, the convergence is not uniform.